# LAB 21 – Collaborative AI Coding: Pair Programming and Version Control Integration

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## TASK1:

Prompt given to AI:

"Generate a simple Python To-Do List Manager with options to add tasks, view tasks, delete tasks and exit. Then improve the program by adding error handling, input validation and a feature to mark tasks as completed.

## Old Code:

```
tasks = []
while True:
    print("\n===== TO-DO LIST MANAGER =====")
    print("1. Add Task")
    print("2. View Tasks")
    print("3. Delete Task")
    print("4. Exit")

    choice = input("Enter your choice: ")
```

```
    if choice == "1":
        task = input("Enter task: ")
        tasks.append(task)
        print("Task added successfully!")

    elif choice == "2":
        for i, task in enumerate(tasks, start=1):
            print(i, ".", task)

    elif choice == "3":
        task_no = int(input("Enter task number: "))
        tasks.pop(task_no - 1)
        print("Task deleted successfully!")

    elif choice == "4":
        break

    else:
        print("Invalid choice!")
```

## IMPROVED CODE:

```
tasks = []

while True:
    print("\n===== TO-DO LIST MANAGER =====")
    print("1. Add Task")
    print("2. View Tasks")
```

```
print("3. Delete Task")
print("4. Exit")

choice = input("Enter your choice: ")

if choice == "1":
    task = input("Enter task: ")
    tasks.append(task)
    print("Task added successfully!")

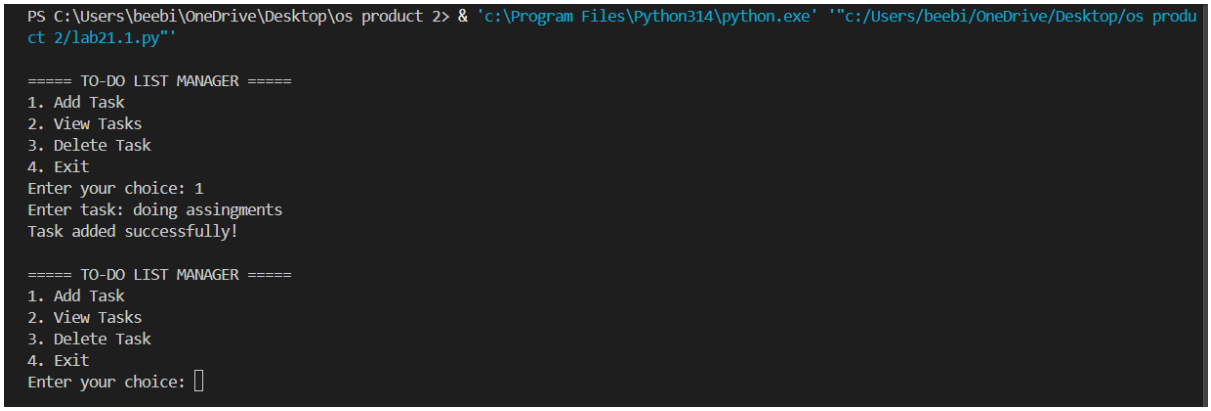
elif choice == "2":
    for i, task in enumerate(tasks, start=1):
        print(i, ".", task)

elif choice == "3":
    if tasks:
        task_no = int(input("Enter task number: "))
        if 1 <= task_no <= len(tasks):
            tasks.pop(task_no - 1)
            print("Task deleted successfully!")
        else:
            print("Invalid task number!")
    else:
        print("No tasks to delete!")

elif choice == "4":
    break
```

```
    print("Invalid choice!")
```

## OUTPUT:




## TASK2:

## Prompt given to AI:

"Explain how to initialize a Git repository, create and switch to a new branch, add AI-assisted code modifications, commit changes with a meaningful message and merge the changes into the main branch. Provide the Git commands with explanations."

## Old Code:

```
tasks = []


while True:
    print("\n===== TO-DO LIST MANAGER =====")
    print("1. Add Task")
    print("2. View Tasks")
    print("3. Delete Task")
    print("4. Exit")

    choice = input("Enter your choice: ")
```

```
    if choice == "1":
        task = input("Enter task: ")
        tasks.append(task)
        print("Task added successfully!")

    elif choice == "2":
        for i, task in enumerate(tasks, start=1):
            print(i, ".", task)

    elif choice == "3":
        task_no = int(input("Enter task number: "))
        tasks.pop(task_no - 1)
        print("Task deleted successfully!")

    elif choice == "4":
        break

    else:
        print("Invalid choice!")
```

## IMPROVED CODE:

```
tasks = []

def add_task():
    task = input("Enter task: ").strip()

    if task:
```

```
        tasks.append({"name": task, "completed": False})
        print("Task added successfully!")
    else:
        print("Task cannot be empty!")


def view_tasks():
    if not tasks:
        print("No tasks available!")
        return

    for i, task in enumerate(tasks, start=1):
        status = "Completed" if task["completed"] else "Pending"
        print(i, task["name"], "-", status)


def delete_task():
    view_tasks()

    if not tasks:
        return

    try:
        n = int(input("Enter task number: "))

        if 1 <= n <= len(tasks):
            tasks.pop(n - 1)
```

```
            print("Task deleted successfully!")
        else:
            print("Invalid task number!")

    except ValueError:
        print("Enter a valid number!")



def complete_task():
    view_tasks()

    if not tasks:
        return

    try:
        n = int(input("Enter task number: "))

        if 1 <= n <= len(tasks):
            tasks[n-1]["completed"] = True
            print("Task completed!")
        else:
            print("Invalid task number!")

    except ValueError:
        print("Enter a valid number!")
```

```
while True:
    print("\n===== TO-DO LIST MANAGER =====")
    print("1. Add Task")
    print("2. View Tasks")
    print("3. Delete Task")
    print("4. Complete Task")
    print("5. Exit")

    choice = input("Enter choice: ")

    if choice == "1":
        add_task()
    elif choice == "2":
        view_tasks()
    elif choice == "3":
        delete_task()
    elif choice == "4":
        complete_task()
    elif choice == "5":
        break
    else:
        print("Invalid choice!")
```

## OUTPUT:

```
PS C:\Users\beebi\OneDrive\Desktop\ai assisted coding> & 'C:\Program Files\Python314\python.exe' '"c:/Users/beebi/OneDrive/Desktop/ai assisted coding/lab21.1.2.py"'

===== TO-DO LIST MANAGER =====
1. Add Task
2. View Tasks
3. Delete Task
4. Complete Task
5. Exit
Enter choice: 2
No tasks available!

===== TO-DO LIST MANAGER =====
1. Add Task
2. View Tasks
3. Delete Task
4. Complete Task
5. Exit
Enter choice: 1
Enter task: write
Task added successfully!
```

## Git Commands:

```
git init
git add .
git commit -m "Add original todo manager"

git branch feature/improvements
git switch feature/improvements

git add .
git commit -m "Add AI-assisted improvements"

git switch main
git merge feature/improvements

git log --oneline --all --graph
```

## TASK3:

### Prompt given to AI:

"Explain how to initialize a Git repository, create and switch to a new branch, add AI-assisted code modifications, commit changes with a meaningful message and merge the changes into the main branch. Provide the Git commands with explanations."

## Old Code:

tasks = []

task = input("Enter task: ")

```
tasks.append(task)

print("Tasks:", tasks)

task_no = int(input("Enter task number to delete: "))
tasks.pop(task_no - 1)

print("Updated tasks:", tasks)
```

## IMPROVED CODE:

```
tasks = []

def add_task():
    task = input("Enter task: ").strip()

    if task:
        tasks.append({"name": task, "completed": False})
        print("Task added")
    else:
        print("Task cannot be empty")

def view_tasks():
    for i, task in enumerate(tasks, start=1):
        status = "Completed" if task["completed"] else "Pending"
        print(i, task["name"], "-", status)

def complete_task():
```

```
    view_tasks()

    try:
        number = int(input("Enter task number: "))

        if 1 <= number <= len(tasks):
            tasks[number - 1]["completed"] = True
            print("Task completed")
        else:
            print("Invalid task number")

    except ValueError:
        print("Enter a valid number")

add_task()
view_tasks()
complete_task()
view_tasks()
```

## OUTPUT:

```
PS C:\Users\beebi\OneDrive\Desktop\ai assisted coding> & 'c:\Program Files\Python314\python.exe' '"c:/Users/beebi/OneDrive/Desktop/ai assisted coding/lab21.3.py"'
Enter task: 2
Task added
1 2 - Pending
1 2 - Pending
Enter task number: 5
Invalid task number
1 2 - Pending
PS C:\Users\beebi\OneDrive\Desktop\ai assisted coding>
```

## Git Commands:

```
git add .
git commit -m "Improve todo manager with validation and task completion"

git log --oneline
```

## TASK4:

### Prompt given to AI:

"Create a simple Python Calculator program. Student A writes the basic calculator with addition, subtraction, multiplication and division. Student B uses AI assistance to improve the program by adding input validation, error handling and better code organization using functions. Explain how both students can collaborate using Git branches and merging."

## Old Code:

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = int(input("Enter choice: "))

if choice == 1:
    print("Result:", num1 + num2)
elif choice == 2:
    print("Result:", num1 - num2)
elif choice == 3:
    print("Result:", num1 * num2)
elif choice == 4:
    print("Result:", num1 / num2)
else:
```

```
    print("Invalid choice")
```

## IMPROVED CODE:

```
def calculator():
    try:
        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        print("\n1. Addition")
        print("2. Subtraction")
        print("3. Multiplication")
        print("4. Division")

        choice = input("Enter choice: ")

        if choice == "1":
            result = num1 + num2
        elif choice == "2":
            result = num1 - num2
        elif choice == "3":
            result = num1 * num2
        elif choice == "4":
            if num2 == 0:
                print("Cannot divide by zero")
                return
            result = num1 / num2
        else:
```

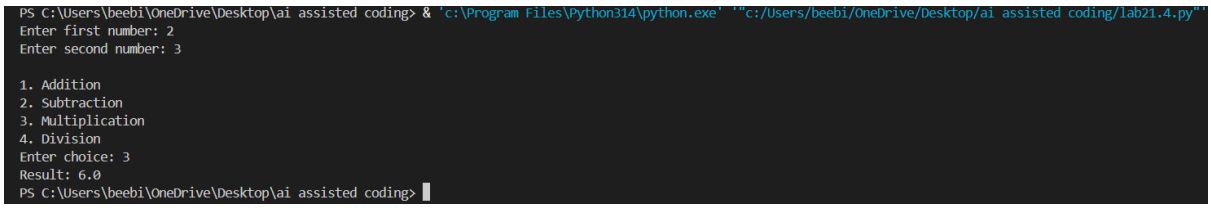
```
            print("Invalid choice")
            return

        print("Result:", result)

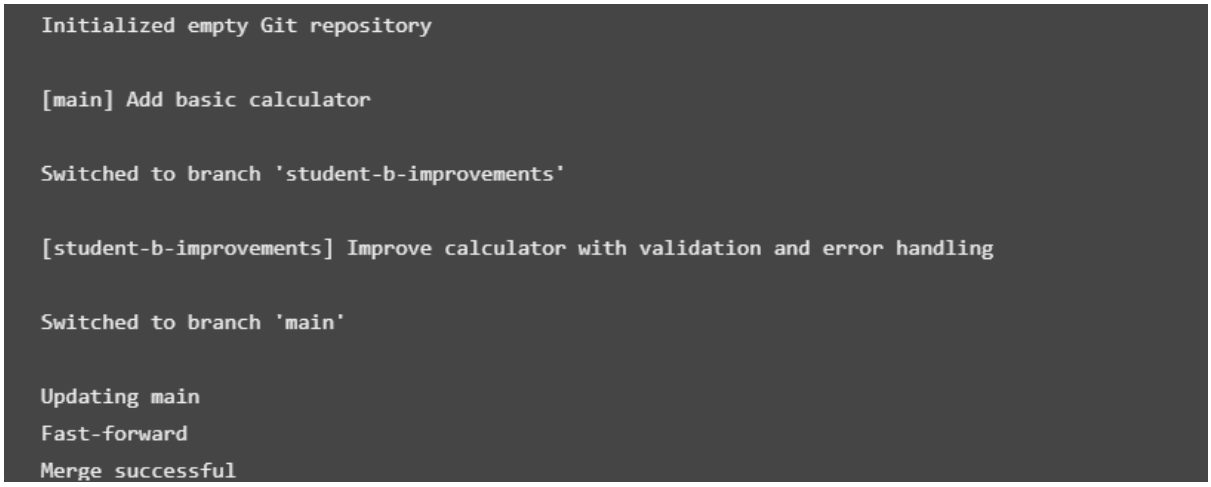
    except ValueError:
        print("Please enter valid numbers")

calculator()
```

## OUTPUT:




## Git Commands:




## TASK5:

### Prompt given to AI:

"Find and fix the syntax and logical errors in the following Python program. Explain the errors and provide the corrected code."

## Old Code:

```
# Program to calculate the average of three numbers

num1 = input("Enter first number: ")
num2 = input("Enter second number: ")
num3 = input("Enter third number: ")

total = num1 + num2 + num3
average = total / 2

if average > 50
    print("Pass")
else:
    print("Fail")
```

## IMPROVED CODE:

```
# Program to calculate the average of three numbers

try:
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    num3 = float(input("Enter third number: "))

    total = num1 + num2 + num3
    average = total / 3

    print("Total:", total)
    print("Average:", average)
```

```
    if average >= 50:
        print("Pass")
    else:
        print("Fail")

except ValueError:
    print("Please enter valid numbers")
```

## OUTPUT:




## COMPARISION:

| Buggy Version | Corrected Version |
|---|---|
| Missing colon after if | Added the missing colon |
| Used input() without converting to numbers | Used float() |
| Calculated average by dividing by 2 | Corrected division to 3 |
| No invalid-input handling | Added try-except |
| Incorrect pass condition (> 50) | Uses >= 50 |

## TASK6:

### Prompt given to AI:

"Create a basic Python calculator with addition, subtraction, multiplication and division. Use AI to improve it by adding calculation history, input validation, division-by-zero handling and better code organization."

## Old Code:

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))


print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")


choice = int(input("Enter choice: "))


if choice == 1:
    print("Result:", num1 + num2)
elif choice == 2:
    print("Result:", num1 - num2)
elif choice == 3:
    print("Result:", num1 * num2)
elif choice == 4:
    print("Result:", num1 / num2)
else:
    print("Invalid choice")
```

## IMPROVED CODE:

```
history = []

def calculator():
    try:
        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        print("\n1. Addition")
        print("2. Subtraction")
        print("3. Multiplication")
        print("4. Division")

        choice = input("Enter choice: ")

        if choice == "1":
            result = num1 + num2
            operator = "+"
        elif choice == "2":
            result = num1 - num2
            operator = "-"
        elif choice == "3":
            result = num1 * num2
            operator = "*"
        elif choice == "4":
            if num2 == 0:
                print("Cannot divide by zero")
                return
```

```
            result = num1 / num2
            operator = "/"
        else:
            print("Invalid choice")
            return

        print("Result:", result)
        history.append(f"{num1} {operator} {num2} = {result}")

    except ValueError:
        print("Please enter valid numbers")

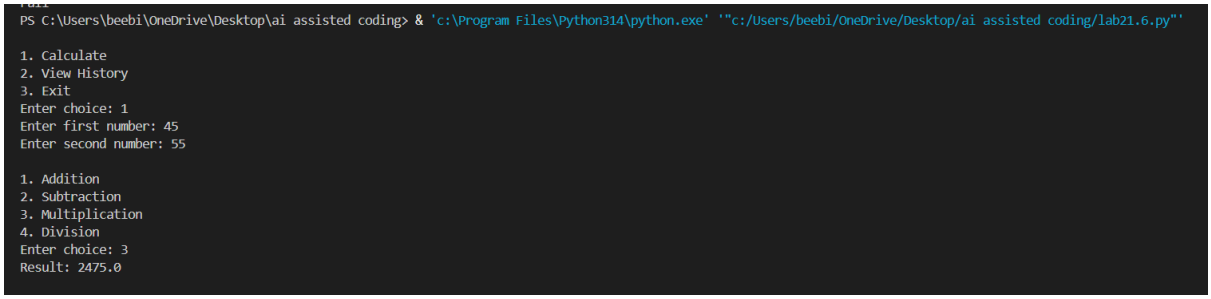
def show_history():
    if not history:
        print("No calculation history")
    else:
        print("\nCalculation History:")
        for item in history:
            print(item)

while True:
    print("\n1. Calculate")
    print("2. View History")
    print("3. Exit")

    choice = input("Enter choice: ")
```

```
if choice == "1":
    calculator()
elif choice == "2":
    show_history()
elif choice == "3":
    print("Exiting calculator")
    break
else:
    print("Invalid choice")
```

# OUTPUT:




## Analysis of Improvements

| Basic Calculator | AI-Enhanced Calculator |
|---|---|
| Performs basic calculations | Performs calculations and stores history |
| No input validation | Handles invalid inputs |
| No division-by-zero protection | Prevents division by zero |
| No history feature | Displays previous calculations |
| Limited functionality | More user-friendly and reliable |